

FACULTY OF ENGINEERING
DIPLOMA ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY



Introduction to Machine Learning (BE103711)

QUESTION BANK mid exam

5 MARKS QUESTIONS (15 Questions)

UNIT 1: Introduction to Operating System

Important 6 Marks Questions

1. Explain the fundamental goals and services of an Operating System with a neat diagram.
2. Compare Multiprogramming, Time Sharing, Real-Time, Multithreading, and Distributed Operating Systems.
3. Explain the architecture and features of the Linux Operating System.
4. Describe the generations of Operating Systems with suitable examples.
5. Compare Linux and the latest Windows Operating System based on architecture, features, security, and applications.

UNIT 2: Process Management and Inter Process Communication

Important 6 Marks Questions

1. Explain the Process Life Cycle with a neat diagram. Describe the role of the Process Control Block (PCB).
2. Explain FCFS, Shortest Job First (SJF), and Round Robin Scheduling Algorithms with suitable examples.
3. Discuss CPU Scheduling Criteria (CPU Utilization, Throughput, Turnaround Time, Waiting Time, and Response Time).
4. Explain Inter Process Communication (IPC), Critical Section Problem, Semaphore, and Race Condition with examples.
5. Explain Deadlock. Discuss its characteristics, prevention, avoidance, detection, and recovery techniques.

UNIT 3: Memory Management

Important 6 Marks Questions

1. Explain Logical Address and Physical Address. Describe memory relocation and protection mechanisms with a neat diagram.
2. Explain Contiguous Memory Allocation. Differentiate between Fixed and Variable Partitioning.
3. Explain Internal Fragmentation, External Fragmentation, and Compaction with suitable examples.
4. Explain First Fit, Best Fit, and Worst Fit Memory Allocation techniques with examples.
5. Explain Paging and Segmentation with neat diagrams. Compare Paging and Segmentation.

MULTIPLE CHOICE QUESTIONS (30 Questions)

UNIT 1: Introduction to Operating System

1. Which of the following is the primary goal of an Operating System?

- A) To increase hardware cost
- B) To manage computer resources efficiently
- C) To write application software
- D) To replace the CPU

✓ Answer: B) To manage computer resources efficiently

2. Which operating system allows multiple users to share CPU time simultaneously?

- A) Batch Operating System
- B) Time Sharing Operating System
- C) Single User Operating System
- D) Embedded Operating System

✓ Answer: B) Time Sharing Operating System

3. Which of the following operating systems is open source?

- A) Windows 11
- B) macOS
- C) Linux
- D) MS-DOS

✓ Answer: C) Linux

4. Which type of operating system is used in air traffic control and medical monitoring systems?

- A) Distributed OS
- B) Time Sharing OS
- C) Real-Time Operating System
- D) Batch OS

✓ Answer: C) Real-Time Operating System

5. Which generation of operating systems introduced multiprogramming?

- A) First Generation
- B) Second Generation
- C) Third Generation
- D) Fourth Generation

✓ Answer: C) Third Generation

UNIT 2: Process Management & Inter Process Communication

1. Which data structure stores information about a process?

- A) Stack
 - B) Queue
 - C) Process Control Block (PCB)
 - D) Buffer
 - ✓ Answer: C) Process Control Block (PCB)
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2. Which CPU scheduling algorithm is non-preemptive?

- A) Round Robin
 - B) FCFS
 - C) Priority (Preemptive)
 - D) Multilevel Queue
 - ✓ Answer: B) FCFS
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3. Which scheduling algorithm uses a fixed time quantum?

- A) SJF
 - B) FCFS
 - C) Round Robin
 - D) Priority Scheduling
 - ✓ Answer: C) Round Robin
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4. Which synchronization mechanism is commonly used to solve the critical section problem?

- A) Cache
 - B) Semaphore
 - C) Stack
 - D) Buffer
 - ✓ Answer: B) Semaphore
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5. Which of the following is one of Coffman's deadlock conditions?

- A) Mutual Exclusion
 - B) Paging
 - C) Segmentation
 - D) Swapping
 - ✓ Answer: A) Mutual Exclusion
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UNIT 3: Memory Management

1. Which memory address is generated by the CPU?

- A) Physical Address
 - B) Logical Address
 - C) Virtual Disk Address
 - D) Cache Address
 - ✓ Answer: B) Logical Address
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2. Which memory allocation technique selects the first available free partition?

- A) Best Fit
- B) Worst Fit
- C) First Fit
- D) Next Fit

☒ Answer: C) First Fit

3. Which type of fragmentation occurs in fixed partition memory allocation?

- A) External Fragmentation
- B) Internal Fragmentation
- C) Paging
- D) Swapping

☒ Answer: B) Internal Fragmentation

4. Paging is mainly used to eliminate:

- A) Internal Fragmentation
- B) External Fragmentation
- C) Deadlock
- D) Starvation

☒ Answer: B) External Fragmentation

5. Which memory management technique divides memory into logical units of variable size?

- A) Paging
- B) Segmentation
- C) Swapping
- D) Buffering

☒ Answer: B) Segmentation
